DLP A02 – k230066

REGRESSION CNN ANALYSIS:

Convolution layer summary

Layer	Kernel size	Filters
Downconv1.0	3x3	32
Downconv2.0	3x3	64
Rfconv.0	3x3	64
Upconv1.0	3x3	32
Upconv2.0	3x3	3
finalconv	3x3	3

Total convolution layers in the regression CNN = 6.

Diagram-style flow:
Input (1x32x32) -> downconv1 (3x3, 32) -> MaxPool -> downconv2 (3x3, 64) -> MaxPool -> rfconv (3x3, 64) -> upconv1 (3x3, 32) -> Upsample -> upconv2 (3x3, 3) -> Upsample -> finalconv (3x3, 3)

Training epochs

- REG_EPOCHS used: 10
- Notes on training duration: The model converged quickly in first 2-3 epochs and then improved slowly. Training loss decreased from 0.1966 (epoch 1) to 0.0113 (epoch 10), and the curve became almost flat near the end.

Training variations

- Epoch variants tested: EPOCH_VARIANTS was set to [5, 10, 20], but RUN_VARIANTS remained False in the submitted run.
- Loss trend observations: From the available 10-epoch run and regression_loss.png, the major drop is early (epoch 1 to 2), then gradual refinement till epoch 10.
- Output image quality observations: In samples_regression.png, predicted images are smooth and natural-looking but slightly blurry in details. Colors are continuous and realistic, especially background and horse body regions.
- Impact of training duration on performance: Based on observed curve behavior, fewer epochs would likely underfit (still high loss and weak colors), while more epochs could improve sharpness slightly but with diminishing returns after convergence.

CLASSIFICATION VS REGRESSION COMPARISON:

- Validation loss (classification): 1.7850
- Validation accuracy (classification): 36.67%
- Qualitative differences in outputs:
    1. Regression output (samples_regression.png): smoother gradients, more natural transitions, but blurrier boundaries.

2. Classification CNN output (samples_classification.png): sharper edges and segmented color patches, but some blocky artifacts and quantization due to only 24 color bins.

- Did classification improve results? Explain:
    1. Compared to regression visually, classification did not clearly produce more natural images; it produced more discrete/color-binned outputs.
    2. Classification gave a measurable pixel accuracy (36.67%), but this metric is for category prediction and is not directly comparable with regression MSE quality.
    3. In this run, classification is useful for structured color assignment but regression appears more photorealistic for many samples.

SKIP CONNECTIONS (UNet):

- UNet validation loss and accuracy: 1.0651 | 60.33%
- Did skip connections improve results? Why:
    1. Yes. UNet significantly improved over plain classification CNN in this run: validation loss dropped from 1.7850 to 1.0651 and validation accuracy improved from 36.67% to 60.33%.
    2. The unet_loss.png curve also decreases faster and reaches a lower final value than classification_loss.png.
    3. Visually (unet/samples_classification.png), the UNet output is less noisy and preserves object structure better.
- At least two reasons skip connections help:
    I. Skip connections pass low-level spatial details (edges, textures, location cues) directly from encoder to decoder, so decoder does not need to relearn them.
    II. They improve gradient flow during backpropagation, making deeper networks easier to optimize and reducing vanishing-gradient issues.
    III. For pixel-wise tasks like colorization, combining local fine details with global semantic context directly improves color placement consistency.

ACTIVATIONS ANALYSIS:

Requirement note:

- As instructed in the assignment, the visualization block from colourization.ipynb should be run for activation interpretation.
- The explanations below are aligned with activation outputs obtained in the executed Assignment notebook and observed model outputs.

CNN activations:

- Early layer activations: downconv1 has mean 0.6244 and std 0.7289 with maximum 6.5648, showing strong response to basic intensity/edge patterns.
- Deeper layer activations: downconv2 and rfconv have lower maxima (5.2121 and 4.4502), and rfconv mean drops to 0.4203. This suggests deeper features become more selective and compressed, focusing on semantic structure rather than raw edges.

UNet activations:

- Comparison with CNN: UNet activations show a progressive reduction in mean from down1 (0.4833) to bottleneck (0.3171), similar to feature abstraction trend, but with skip pathways this information is not lost for reconstruction.
- Detailed analysis:
    I. CNN without skips relies mostly on compressed deep features for reconstruction, which can cause coarser output.
    II. UNet keeps both shallow and deep information available at decoding stage, so object boundaries and regions are more coherent.
    III. This is consistent with observed improvements in validation metrics and sample quality.

1) Evaluation metrics (Johnson et al., 2016)

How can evaluation be improved to match human judgment?

1) Pixel-wise losses (MSE/CE) are not enough because small spatial shifts get heavily penalized even if image looks fine.
2) A better approach is to include perceptual loss using features from a pretrained network (as in Johnson et al., 2016). Instead of only comparing pixels, compare high-level feature representations.
3) We can also report SSIM/LPIPS-style perceptual quality and perform human preference evaluation (pairwise comparison).
4) Practical setup: total loss = alpha * pixel loss + beta * perceptual loss, then validate with both numeric metrics and visual inspection.

2) Image size generalization (Ronneberger et al., 2015)

How to adapt models trained on 32x32 for larger images?

1) Because these models are convolutional, they can be run on larger images directly if architecture is fully convolutional (no fixed-size fully connected layers).
2) UNet-style encoder-decoder naturally supports multi-scale processing and is suitable for larger resolutions.
3) For memory efficiency, large images can be colorized using overlapping tiles and blended at boundaries.
4) Fine-tuning on larger-resolution samples further improves color consistency and sharpness.
5) Skip connections are especially useful at larger resolutions because they preserve fine spatial detail across scales.